

Predicting Customer Response to Marketing Campaigns Using Classification Models

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Abstract

This project focuses on predicting whether a customer will respond to a marketing campaign using classification techniques. The problem is treated as a supervised learning task because the dataset includes a labeled target variable, Response, which indicates whether a customer accepted or rejected a campaign offer. By using historical customer data, the goal is to build models that can accurately classify future customer responses.

The dataset used in this analysis is the marketing campaign dataset from Kaggle, which includes customer demographic information, purchasing behavior, and interaction history with previous campaigns. Key variables include income, education, marital status, and various spending categories. These features are used as predictors to understand patterns in customer behavior and their likelihood of responding to marketing efforts.

Three classification models were developed and compared: Logistic Regression, Random Forest, and Neural Network. The results show that the Neural Network model performed the best overall, as it provided a better balance between sensitivity and specificity compared to the other models. The findings highlight how customer characteristics and purchasing behavior can be used to improve targeted marketing strategies and increase campaign success rates.

Introduction

This report focuses on analyzing a marketing campaign dataset to predict whether a customer will respond to a marketing offer. The dataset used in this analysis is the Marketing Campaign dataset obtained from Kaggle. It contains information about customers such as demographic characteristics, purchasing behavior, and past interactions with marketing campaigns. The response variable used in this analysis is Response, which indicates whether a customer accepted a campaign offer (1) or not (0). The predictor variables include attributes such as income, education, marital status, number of purchases, and spending across different product categories.

The motivation for this analysis comes from the importance of marketing efficiency in business decision-making. Companies often spend significant resources on marketing campaigns, but not all customers respond to these efforts. A response model can provide a significant boost to the efficiency of a marketing campaign by increasing responses or reducing unnecessary expenses. The objective of this analysis is to predict which customers are most likely to respond to an offer for a product or service.

This problem is appropriate to treat as a supervised learning task because the dataset contains a labeled outcome variable, Response. Supervised learning methods use historical labeled data to learn patterns between predictor variables and the response variable. These patterns can then be

used to make predictions on new data. Classification techniques are commonly used in marketing analytics to improve targeting strategies (Witten et al., 2016).

The primary analytical question guiding this study is: Can customer demographic and purchasing behavior data be used to accurately predict whether a customer will respond to a marketing campaign? To answer this question, three classification algorithms are applied: Logistic Regression, Random Forest, and Neural Network. These methods were selected because they represent different approaches to classification and allow for a meaningful comparison of model performance.

The experimental setup used in this analysis follows the holdout method, where the dataset is divided into training and testing subsets. All models are trained using the same training strategy to ensure a fair comparison. The remainder of this report includes the Data and Methods section, Results section, and Conclusion and Discussion.

Data and Data Preparation

Data Description

The dataset used in this analysis is the Marketing Campaign dataset obtained from Kaggle. It contains customer-level data collected from a marketing campaign, including demographic information, purchasing behavior, and previous campaign responses.

The response variable is Response, which indicates whether a customer accepted a marketing offer (1 = yes, 0 = no). Predictor variables include income, education level, marital status, number of purchases, and spending across different product categories.

The dataset contains both numeric and categorical variables. Numeric variables include income and spending amounts, while categorical variables include education and marital status. These variables provide valuable insight into customer behavior and are used to build classification models.

Attribute	Description	Data Type	Range/Categories
Response	Indicates whether a customer accepted campaign	Categorical	0(no), 1(yes)
Year_Birth	Year customer was	Numeric	~1890–2000

	born		
Education	Education level of customer	Categorical	Basic, Graduation, Master, PhD
Marital_Status	Marital status of customer	Categorical	Single, Married, Divorced, etc.
Income	Customer yearly income	Numeric	0 – 600,000+
Kidhome	Number of children in household	Numeric	0 – 2
Teenhome	Number of teenagers in household	Numeric	0 – 2
MntWines	Amount spent on wine	Numeric	0 – 1,500+
MntFruits	Amount spent on fruits	Numeric	0 – 200+
MntMeatProducts	Amount spent on meat products	Numeric	0 – 1,500+
MntFishProducts	Amount spent on fish products	Numeric	0 – 300+
MntSweetProducts	Amount spent on sweets	Numeric	0 – 300+
MntGoldProds	Amount spent on gold products	Numeric	0 – 400+
NumDealsPurchases	Number of purchases	Numeric	0 – 15

	with discount		
NumWebPurchases	Number of online purchases	Numeric	0 – 30
NumCatalogPurchases	Number of catalog purchases	Numeric	0 – 30
NumStorePurchases	Number of store purchases	Numeric	0 – 15
NumWebVisitsMonth	Number of website visits per month	Numeric	0 – 20

Table 1: provides a description of the main attributes included in the dataset

Data Preparation

Several preprocessing steps were applied before modeling. First, the dataset was imported into R and summary statistics were examined to understand the structure of the data. Observations with missing values were removed to ensure consistency across all models.

Next, variables that do not contribute to prediction were removed, including ID, Dt_Customer, Z_CostContact, and Z_Revenue. These variables either serve as identifiers or contain constant values and do not provide meaningful information for classification.

The response variable Response was converted into a factor variable for classification. Categorical variables such as Education and Marital_Status were also treated as factor variables. In addition, rare categories in marital status were simplified to avoid issues such as zero variance during model training.

These preprocessing steps ensured that the dataset was clean, consistent, and suitable for classification analysis.

Methods

Exploratory Data Analysis

Exploratory data analysis was conducted to understand the structure of the dataset and identify potential issues that may affect model performance. Summary statistics were used to examine distributions of variables such as income and spending behavior.

The distribution of the response variable revealed a strong class imbalance, with approximately 85% of customers not responding and only about 15% responding to the campaign. This imbalance is important because it can influence classification performance and lead to biased models.

Classification Models

Three classification algorithms were used in this analysis:

Logistic Regression is a statistical model used for binary classification that estimates the probability of a customer responding to a campaign. It assumes a linear relationship between the predictors and the log-odds of the response variable.

Random Forest is an ensemble method that builds multiple decision trees and combines their predictions. It is capable of capturing complex relationships in the data and is less prone to overfitting compared to a single decision tree.

Neural Network is a non-linear model that learns patterns through interconnected layers of nodes. It is capable of modeling complex relationships between variables and can provide strong predictive performance when properly trained.

Experimental Setup and Training Strategy

The dataset was partitioned using the holdout method, with 70% of the data used for training and 30% used for testing. A random seed was set to ensure reproducibility.

All models were trained using the same strategy to ensure a fair comparison. Cross-validation was used for the neural network, while the same training setup was applied consistently across models.

Model Evaluation

Model performance was evaluated using confusion matrices and several metrics, including accuracy, error rate, sensitivity (recall), specificity, precision, and F1-score.

Accuracy measures overall correctness, while sensitivity measures the ability to correctly identify non-responders. Specificity measures how well the model identifies actual responders.

Precision and F1-score provide additional insight into model performance, especially in the presence of class imbalance.

Results

Classification Model Performance

Three classification models were developed and evaluated in this analysis: Logistic Regression, Random Forest, and Neural Network. Each model was trained using the same training dataset and evaluated on the testing dataset using confusion matrices and multiple evaluation metrics, including accuracy, sensitivity (recall), specificity, precision, and F1-score.

The distribution of the response variable shows that approximately 85% of customers did not respond to the campaign, while only about 15% did respond. This indicates a strong class imbalance in the dataset. Because most customers do not respond, models may appear to perform well simply by predicting the majority class. Therefore, metrics such as specificity and balanced accuracy are especially important for evaluating how well the models identify customers who actually respond.

Logistic Regression Results

The Logistic Regression model achieved an accuracy of 0.8899. The sensitivity was very high at 0.9645, indicating that the model is very effective at identifying customers who did not respond to the campaign. However, the specificity was much lower at 0.4646, showing that the model struggles to correctly identify customers who actually responded.

The F1-score was 0.9371, and the Kappa value was 0.4976, indicating moderate overall performance.

From a real-world marketing perspective, this means that the Logistic Regression model is effective at identifying customers who are unlikely to respond, which can help reduce unnecessary marketing costs. However, it is less effective at identifying customers who are likely to respond, which could result in missed opportunities for targeting high-value customers.

Random Forest Results

The Random Forest model achieved an accuracy of 0.8793. It showed extremely high sensitivity at 0.9787, meaning it was very effective at identifying non-responders. However, the specificity was very low at 0.3131, indicating poor performance in identifying customers who responded to the campaign.

The F1-score was 0.9324, and the Kappa value was 0.3806, which is lower than the other models.

In a real-world marketing context, this means that the Random Forest model would classify most customers as non-responders. While this may reduce marketing costs by avoiding outreach to unlikely customers, it also means that many potential responders would be missed. This could significantly reduce the effectiveness of a marketing campaign, as the model fails to identify a large portion of customers who might actually respond.

Neural Network Results

The Neural Network model achieved an accuracy of 0.8854. It provided a better balance between sensitivity (0.9504) and specificity (0.5152) compared to the other models. This indicates that the model is more effective at identifying both responders and non-responders.

Figure 1 shows the cross-validation accuracy of the Neural Network model across different numbers of hidden units and weight decay values.

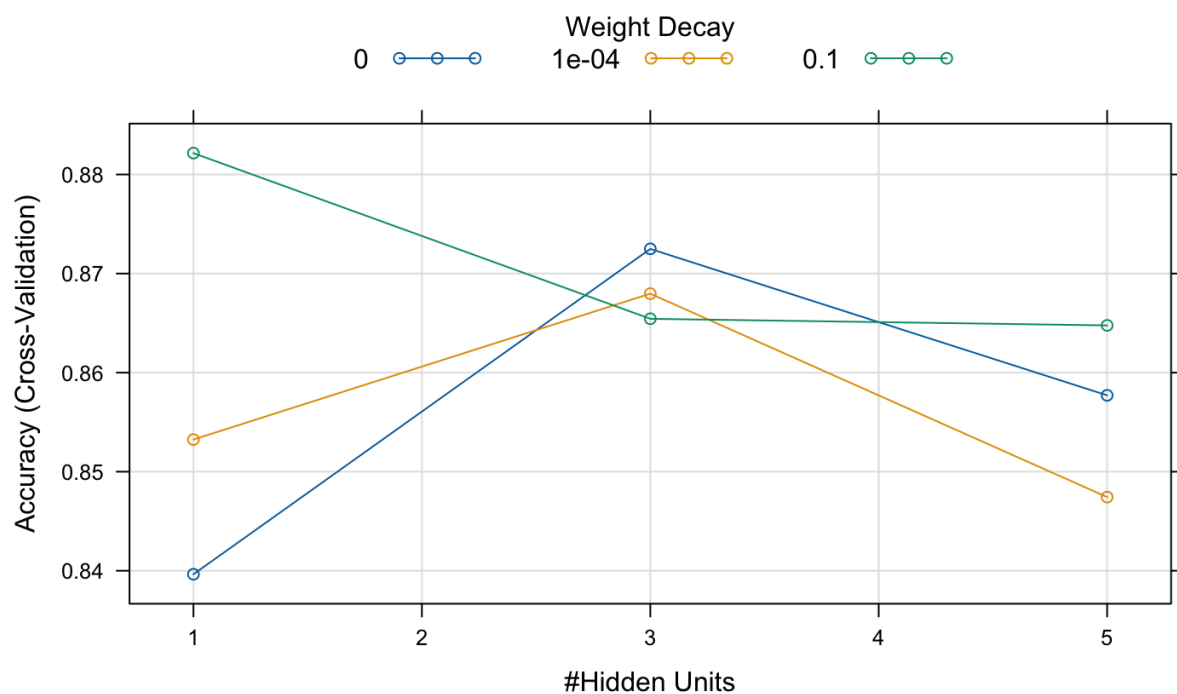


Figure 1: Cross-validation accuracy of Neural Network for different hidden units and weight decay values

As shown in Figure 1, the model achieves its highest performance with a moderate number of hidden units. This suggests that the Neural Network is able to capture complex relationships in the data without overfitting.

The F1-score was 0.9338, and the Kappa value was 0.5078, which is the highest among the three models.

From a marketing perspective, this means that the Neural Network model is more effective at identifying customers who are likely to respond while still maintaining strong performance for non-responders. This balance is important because it allows companies to both reduce unnecessary marketing costs and increase the likelihood of targeting customers who will respond to campaigns.

Model	Accuracy	Sensitivity	Specificity	F1 Score	Kappa
Logistic Regression	0.8899	0.9645	0.4646	0.9371	0.4976
Randon Forest	0.8793	0.9787	0.3131	0.9324	0.3806
Neural Network	0.8854	0.9504	0.51512	0.9338	0.5078

Table 2 summarizes the performance of all three classification models.

Although Logistic Regression achieved the highest overall accuracy, this result is influenced by class imbalance and does not reflect strong performance across both classes. The Random Forest model performed the worst in identifying customers who responded, as shown by its low specificity.

The Neural Network model provided the best overall performance because it achieved the highest Kappa value and the best balance between sensitivity and specificity. This indicates that it is more effective at distinguishing between customers who respond and those who do not.

From a business perspective, this is especially important. A model that only predicts non-responders may reduce costs but fails to identify valuable customers. On the other hand, a model that balances both classes can help companies target the right customers more effectively, increasing campaign success rates and improving return on investment.

Conclusion and Discussion

This analysis focused on predicting whether a customer will respond to a marketing campaign using classification techniques. Three models were applied: Logistic Regression, Random Forest, and Neural Network. The models were trained using a holdout method and evaluated using multiple performance metrics.

The results show that all models achieved high accuracy; however, this was influenced by class imbalance in the dataset. Since most customers did not respond to the campaign, the models were biased toward predicting the majority class.

Among the models, the Neural Network provided the best overall performance because it achieved the highest Kappa value and the best balance between sensitivity and specificity. Logistic Regression also performed well but was limited by its linear nature. Random Forest performed poorly in identifying responders, despite strong performance on the majority class.

One limitation of this analysis is the strong class imbalance, which affects the ability of models to correctly identify the minority class. Another limitation is the use of a single holdout split, which may not fully represent model performance across different data samples.

These limitations affect the interpretation of the results, as high accuracy does not necessarily indicate good performance for identifying responders. Future work could address class imbalance using resampling techniques such as oversampling or SMOTE. Additional models and feature engineering could also improve predictive performance.

Overall, this analysis demonstrates the importance of evaluating classification models beyond accuracy and highlights the challenges of working with imbalanced datasets in marketing analytics.

Reference

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